CHAPTER 5

GEOMORPHIC PROCESSES

CHAPTER ESSENCE AT A GLANCE:
Distinguish between endogenic relief-building forces and exogenic denudational processes, mastering weathering mechanisms, slow and rapid mass movements, erosion, and landscape gradation.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Endogenic versus Exogenic Dualism:** Endogenic forces originating deep within the Earth build relief through diastrophism and volcanism; exogenic forces derive energy from solar insolation and gravity, wearing down elevations through denudation.

• **Weathering Mechanisms:** In situ disintegration and decomposition of rocks without transport; categorized into Physical (frost wedging, thermal expansion, unloading, salt crystal growth), Chemical (solution, carbonation, hydration, oxidation), and Biological.

• **Mass Movements and Gravitational Failure:** Downslope movement of regolith and debris under direct gravitational shear stress; categorized into slow mass movements (soil creep, solifluction) and rapid movements (earthflow, mudflow, debris avalanche, landslides).

• **Denudation and Gradation Equilibrium:** Denudation comprises weathering, mass movements, erosion, and transportation, lowering land surfaces towards the base level of erosion to create a continuous dynamic balance.

ESSENTIAL GEOGRAPHICAL MODELS & LANDFORMS

$$\text{Denudation Total} = \text{Weathering} + \text{Mass Movements} + \text{Erosion} + \text{Transportation}$$

Fundamental Exogenic Denudation Identity Equation

$$F_s = \frac{\text{Shear Strength of Regolith}}{\text{Shear Stress along Slip Plane}} \rightarrow F_s < 1 \rightarrow \text{Slope Failure}$$

Factor of Safety Governing Mass Movement and Landslides

$$CaCO_3 + H_2O + CO_2 \rightleftharpoons Ca(HCO_3)_2 \rightarrow \text{Soluble Bicarbonate}$$

Chemical Carbonation Equilibrium in Karst Weathering

$$\text{Rate of Denudation} = \frac{\Delta \text{Mass Removed}}{\Delta \text{Time} \times \text{Basin Area}}$$

Geomorphic Basin Mass Gradation Flux Equation

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Weathering is strictly an in situ (static) process involving no substantial transportation of debris, whereas erosion is an active dynamic process involving detachment, acquisition, and transportation by mobile geomorphic agents.

• **Alert:** Mass movement does not require a geomorphic agent (like running water, wind, or glacier) to carry debris; gravity is the direct driving force, while water acts only as a lubricant and weight adder.

• **Alert:** Solifluction occurs exclusively in cold permafrost periglacial climates where water-saturated thawed surface soil creeps over an underlying impermeable permanently frozen layer (permafrost).

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Distinguish between Endogenic and Exogenic geomorphic forces with one example of each. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Endogenic Forces:** Internal forces originating within the Earth driven by geothermal energy and radioactivity that elevate relief (e.g. volcanism, folding, faulting). **[1 Mark]**

• **Exogenic Forces:** External forces operating on the Earth surface driven by solar energy and gravity that wear down relief through gradation (e.g. weathering, running water erosion). **[1 Mark]**

2. Define Weathering and explain why it is described as an in situ process. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Definition:** Weathering is the mechanical disintegration and chemical decomposition of rocks under the direct influence of atmospheric elements. **[1 Mark]**

• **In Situ Character:** It is called an in situ (on-site) process because rocks break down and alter at their original resting place with little or no transportation of the weathered debris. **[1 Mark]**

3. Explain three major chemical weathering processes operating in humid tropical environments. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **1. Solution:** Soluble minerals like rock salt and gypsum dissolve directly in water and are leached away in solution. **[1 Mark]**

• **2. Carbonation:** Rainwater absorbs atmospheric carbon dioxide to form weak carbonic acid, which dissolves calcium carbonate rocks into soluble calcium bicarbonate. **[1 Mark]**

• **3. Oxidation:** Chemical combination of dissolved oxygen with iron-bearing minerals in rocks to form iron oxides (rusting), weakening the rock structure. **[1 Mark]**

4. Distinguish between slow mass movements (creep) and rapid mass movements (landslides). [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Slow Mass Movements (Creep):** Extremely slow, continuous downslope movement of soil or regolith, imperceptible except through bent telephone poles, tilted fences, or curved tree trunks. **[1.5 Marks]**

• **Rapid Mass Movements (Landslides):** Rapid, sudden, and catastrophic sliding or falling of large masses of rock and regolith along distinct failure planes under gravitational stress. **[1.5 Marks]**

5. What is Denudation? Explain the four major processes that collectively bring about the denudation of the Earth's surface. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Denudation Concept:** Denudation literally means to strip off or uncover; it is the collective term for all exogenic processes that wear away the Earth's surface. **[1 Mark]**

• **1. Weathering:** In situ physical disintegration and chemical decomposition of rocks by atmospheric elements without transport. **[0.75 Mark]**

• **2. Mass Movements:** Downslope transfer of weathered rock debris down a slope under direct gravitational force. **[0.75 Mark]**

• **3. Erosion:** Dynamic detachment, acquisition, and removal of rock debris by mobile geomorphic agents such as running water, wind, or glaciers. **[0.75 Mark]**

• **4. Transportation:** Conveyance of eroded rock particles over long distances before final sedimentation in depositional basins. **[0.75 Mark]**

6. Discuss the natural and anthropogenic factors that trigger severe landslides in the Western Ghats and Himalayan regions. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Natural Factors:** Steep topographical slopes, highly fractured and folded tectonic rock strata, heavy torrential monsoon rainfall saturating regolith, and frequent seismic shocks. **[2 Marks]**

• **Anthropogenic Factors:** Extensive deforestation destabilizing root cohesion, unscientific slope cutting for roads and railway tracks, blasting for quarrying, and overloading slopes with construction. **[2 Marks]**